

MAGNITOGORSK AP, Russian Federation

WMO: 232215

Lat: **53.393N**Lon: **58.756E**Elev: **436**StdP: **96.20**Time Zone: **5.00 (E05)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-26.0	-31.1	0.2	-27.5	-29.1	0.3	-25.6	11.4	-11.5	10.4	-11.1	1.9	10	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	30.9	17.4	29.0	17.3	27.1	16.4	19.4	26.9	18.5	25.9	17.7	24.6	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.7	21.9	16.0	12.0	20.9	15.1	11.3	20.3	56.9	26.2	54.2	25.8	51.6	24.4	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.5	8.5	-30.4	33.2	2.9	1.5	-32.5	34.3	-34.3	35.2	-35.9	36.0	-38.0	37.1		
(5)			-30.4	21.4	2.8	1.0	-32.4	22.0	-34.0	22.6	-35.6	23.1	-37.6	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-15.7	-13.6	-6.5	5.1	13.1	17.7	19.1	17.7	11.4	3.6	-4.1	-11.6
	DBStd	13.41	5.90	6.45	5.10	5.20	4.85	4.02	3.37	4.33	4.07	4.98	6.08	6.44
	HDD10.0	3485	796	661	511	164	24	2	0	2	30	205	422	668
	HDD18.3	5709	1054	894	769	397	171	59	31	65	212	457	672	927
	CDD10.0	970	0	0	0	17	121	233	281	241	71	6	0	0
	CDD18.3	153	0	0	0	0	9	41	54	46	3	0	0	0
	CDH23.3	1714	0	0	0	5	164	466	557	487	34	0	0	0
	CDH26.7	489	0	0	0	0	21	137	156	171	5	0	0	0
Wind	WSAvg	3.9	3.4	3.5	4.4	4.5	4.2	4.0	3.7	3.7	3.6	4.2	4.2	3.6
Precipitation	PrecAvg	597	45	36	45	38	47	63	73	59	42	49	49	51
	PrecMax	818	85	78	90	93	91	138	156	157	84	111	89	126
	PrecMin	455	8	6	3	4	4	15	20	25	13	10	4	13
	PrecStd	101	19	17	24	21	24	28	32	28	18	23	23	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	2.2	7.8	24.0	28.9	32.1	33.0	32.9	27.0	19.1	9.2	1.9
		MCWB	-1.9	-0.4	2.4	11.5	15.5	16.3	17.2	18.2	15.2	9.5	6.4	0.1
	2%	DB	-2.8	0.1	3.2	20.1	27.0	30.1	30.1	30.9	23.2	16.0	6.8	-0.1
		MCWB	-3.6	-1.4	0.1	10.5	14.5	16.8	17.7	18.0	13.3	9.3	4.5	-1.3
	5%	DB	-4.1	-2.0	1.9	17.1	25.2	28.1	28.1	28.8	21.1	12.9	4.9	-1.2
		MCWB	-5.0	-3.2	-0.2	9.2	13.5	16.2	17.2	17.8	12.9	7.1	3.1	-2.2
	10%	DB	-6.2	-4.8	0.8	14.0	23.1	26.0	26.0	26.0	18.8	10.2	3.0	-2.8
		MCWB	-7.2	-6.0	-0.7	7.8	12.7	15.9	16.7	17.3	12.0	6.6	1.3	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.2	0.5	2.6	12.8	16.7	19.8	20.2	20.3	16.5	11.3	7.1	-0.2
		MCDB	-1.6	1.6	6.3	21.3	25.8	27.2	27.2	27.5	24.4	16.8	9.0	0.6
	2%	WB	-3.5	-1.4	0.7	11.0	15.4	18.4	19.3	19.4	14.8	9.4	4.7	-1.2
		MCDB	-2.7	-0.2	1.9	18.8	24.7	25.9	27.0	26.9	21.4	14.3	6.0	-0.5
	5%	WB	-5.3	-3.3	0.0	9.7	14.5	17.5	18.5	18.5	13.7	8.2	3.1	-2.7
		MCDB	-4.4	-2.0	1.4	16.0	22.9	24.8	25.7	26.1	19.0	12.0	4.4	-1.7
	10%	WB	-7.4	-5.8	-1.0	8.1	13.4	16.7	17.6	17.6	12.6	7.0	1.4	-3.7
		MCDB	-6.4	-4.7	0.5	13.8	21.4	23.6	24.1	24.8	17.3	9.7	2.6	-2.8
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.3	8.1	10.2	13.1	12.2	11.6	11.2	10.2	7.8	6.3	6.7
		MCDBR	6.4	8.6	8.6	15.1	15.3	15.0	14.7	15.2	13.9	12.4	7.4	6.7
		MCWBR	6.0	7.2	6.8	7.6	6.4	5.8	5.3	5.3	6.3	6.9	5.5	6.0
	5% WB	MCDBR	6.2	8.0	8.1	13.9	14.3	12.4	12.4	13.2	12.0	10.2	6.3	6.5
		MCWBR	5.9	6.7	6.9	7.7	6.4	5.4	5.3	5.3	6.2	6.4	5.2	6.0
Clear-Sky Solar Irradiance	taub	0.272	0.298	0.352	0.401	0.391	0.385	0.406	0.422	0.367	0.335	0.299	0.268	
	taud	2.126	2.096	2.005	2.123	2.274	2.360	2.297	2.229	2.349	2.334	2.234	2.122	
	Ebn at Noon	682	784	809	823	856	866	837	796	802	744	648	604	
	Edh at Noon	68	98	134	137	125	116	121	123	96	78	62	57	
All-Sky Solar Radiation	RadAvg	1.04	2.13	3.64	4.57	5.47	5.98	5.64	4.60	3.20	1.74	0.88	0.69	
	RadStd	0.09	0.15	0.28	0.40	0.47	0.51	0.38	0.44	0.32	0.30	0.11	0.09	